

Physical Limnology of Ponds - Knowledge Gaps and Research Perspectives

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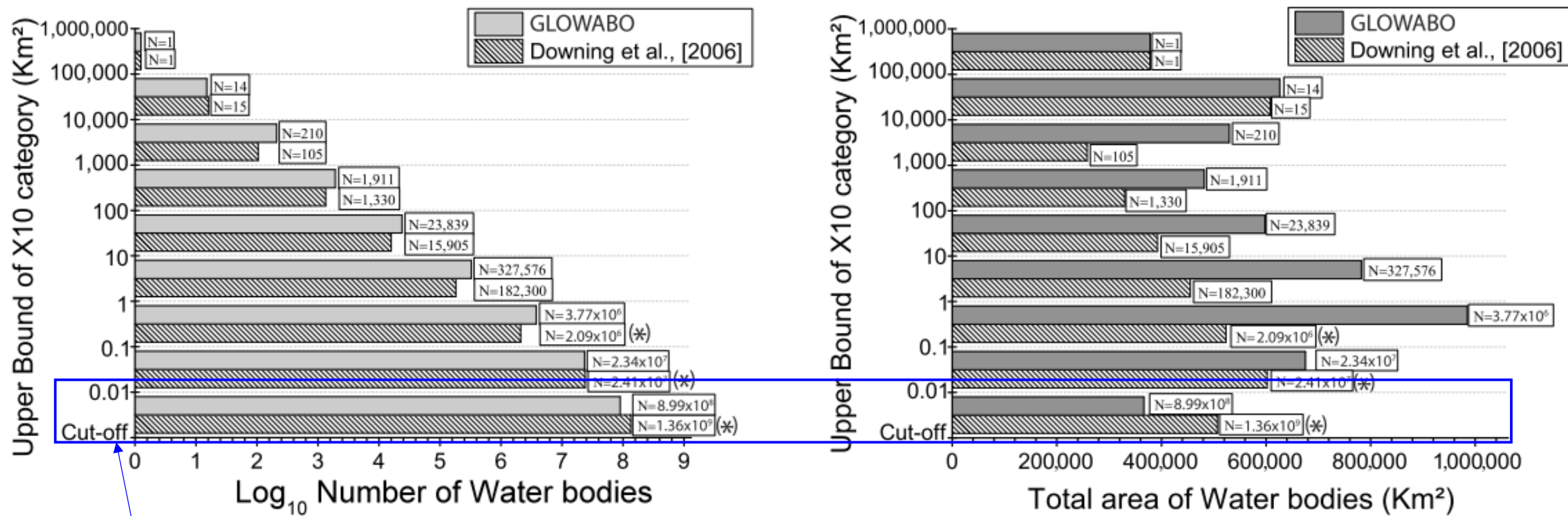
University of Kaiserslautern-Landau (RPTU)



What are ponds and why are they important?

A global inventory of lakes based on high-resolution satellite imagery

Charles Verpoorter^{1,2}, Tiit Kutser^{2,3}, David A. Seekell^{4,5}, and Lars J. Tranvik²

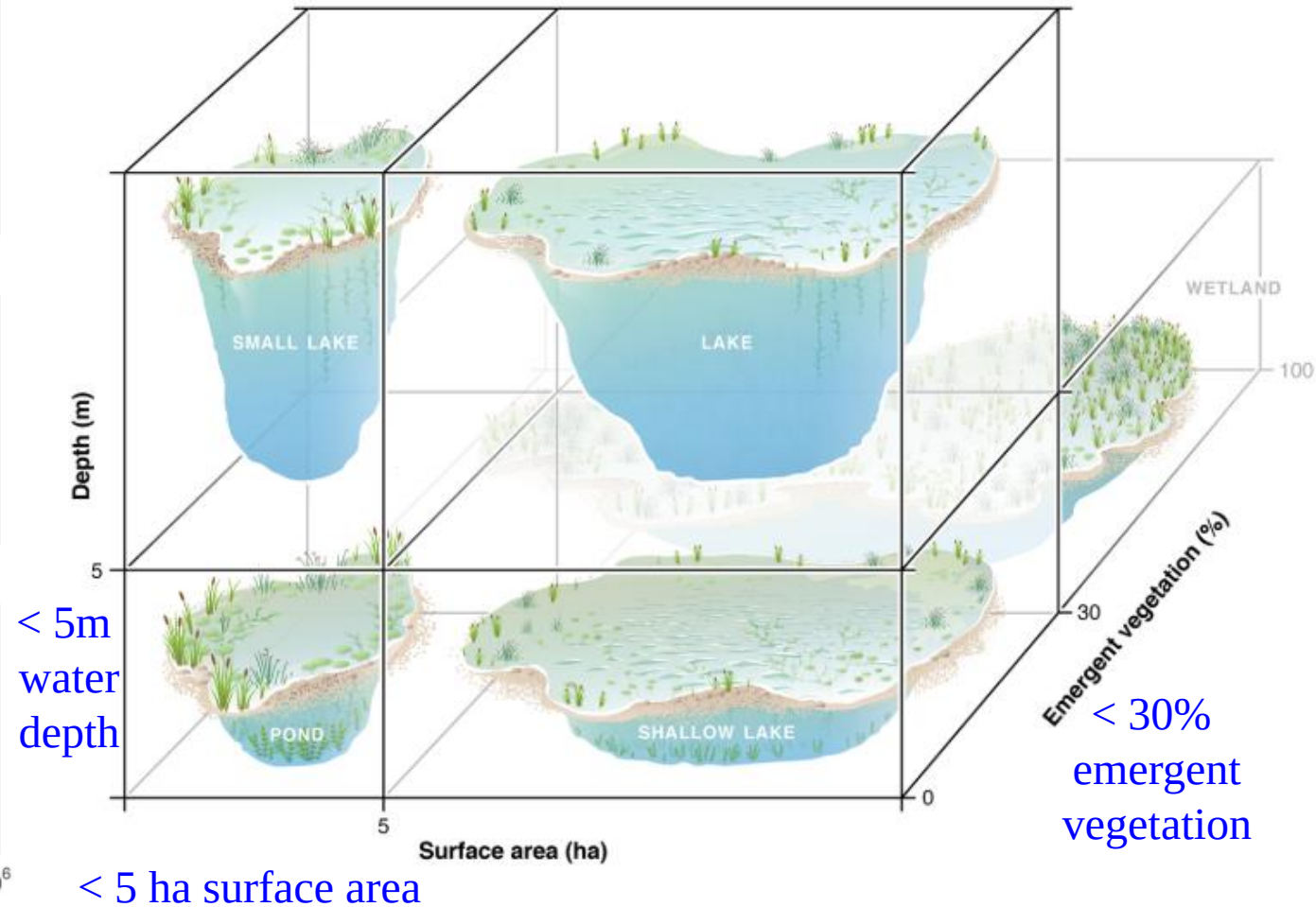
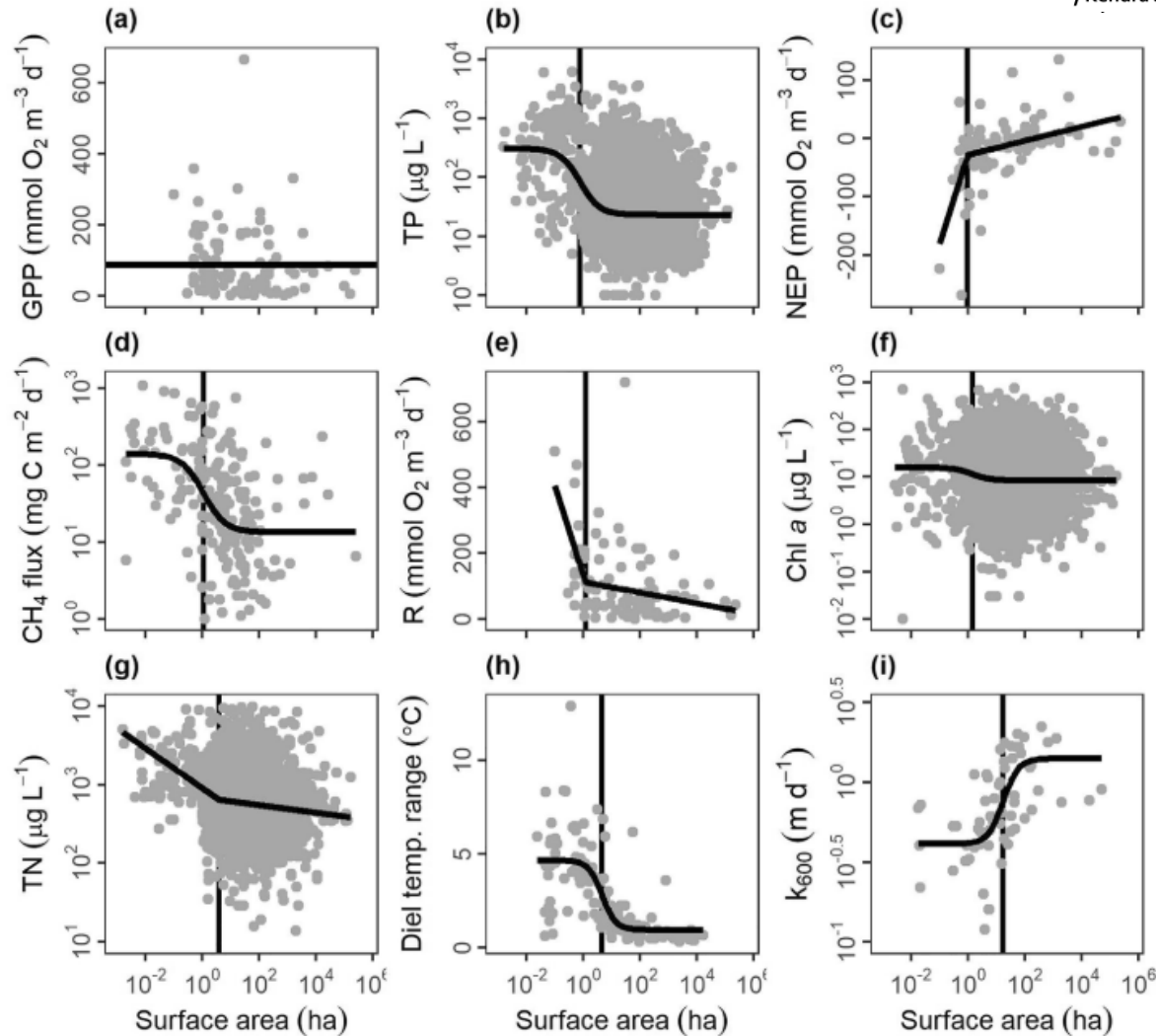


0.1 – 1 ha surface area

What are ponds and why are they important?

A functional definition to distinguish ponds from lakes and wetlands

David C. Richardson^{1,19}, Meredith A. Holgerson^{2,19}, Matthew J. Farragher³, Kathryn K. Hoffman⁴, Katelyn B. S. King⁵, Maria B. Alfonso⁶, Mikkel R. Andersen⁷, Kendra Spence Cheruveil⁸, Kristen A. Coleman⁹, Mary Jade Farruggia¹⁰,



What are ponds and why are they important?

Mapping global lake dynamics reveals the emerging roles of small lakes

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Xuehui Pi^{1,2,3,13}, Qiuqi Luo^{1,13}, Lian Feng¹, Yang Xu^{1,4}, Jing Tang^{5,6},
Xiuyu Liang¹, Enze Ma¹, Ran Cheng⁷, Rasmus Fensholt⁴, Martin Brandt⁴,
Xiaobin Cai⁸, Luke Gibson¹, Junguo Liu^{1,9}, Chunmiao Zheng^{1,10},
Weifeng Li^{2,3,11} & Brett A. Bryan¹²

Small artificial waterbodies are widespread and persistent emitters of methane and carbon dioxide

M. Peacock¹ | J. Audet² | D. Bastviken³ | S. Cook⁴ | C. D. Evans^{1,5} |
A. Grinham⁶ | M. A. Holgerson⁷ | L. Högbom^{8,9} | A. E. Pickard¹⁰ | P. Zieliński¹¹ |
M. N. Futter¹

Extreme diel dissolved oxygen and carbon cycles in shallow vegetated lakes

Mikkel R. Andersen, Theis Kragh and Kaj Sand-Jensen

The importance of small waterbodies for biodiversity and ecosystem services: implications for policy makers

J. Biggs · S. von Fumetti · M. Kelly-Quinn

Biogeochemical hotspots: Role of small water bodies in landscape nutrient processing

Frederick Y. Cheng¹ and Nandita B. Basu^{1,2,3}

High carbon burial rates by small ponds in the landscape

Scott Taylor¹, Peter J Gilbert², David A Cooke¹, Michael E Deary¹, and Michael J Jeffries^{1*}

Eutrophic urban ponds suffer from cyanobacterial blooms: Dutch examples

Guido W. A. M. Waajen · Elisabeth J. Faassen · Miquel Lüring

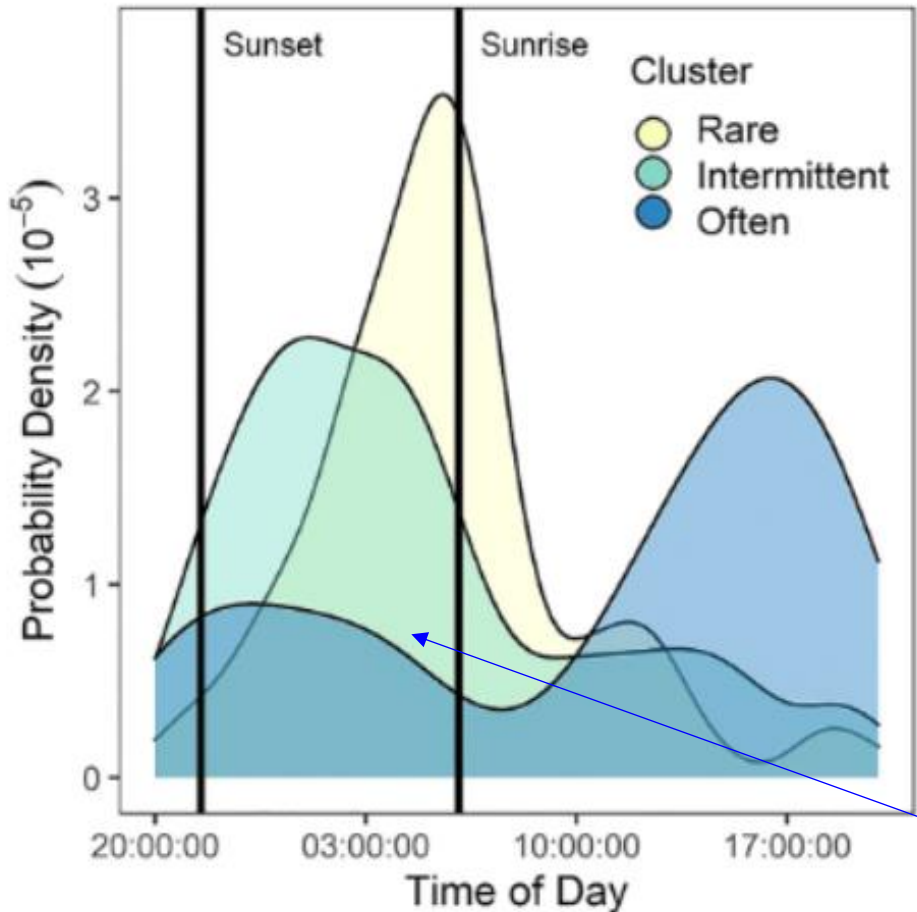
Shallow plant-dominated lakes – extreme environmental variability, carbon cycling and ecological species challenges

Kaj Sand-Jensen^{1,*}, Mikkel René Andersen^{1,2}, Kenneth Thorø Martinsen¹, Jens Borum¹, Emil Kristensen¹ and Theis Kragh¹

Need for including ponds into regional to global scale biogeochemical models (land surface models).
Need for improving pond management and mitigation measures.
→ What hydrodynamic processes regulate the temperature distribution and mixing dynamics in ponds?

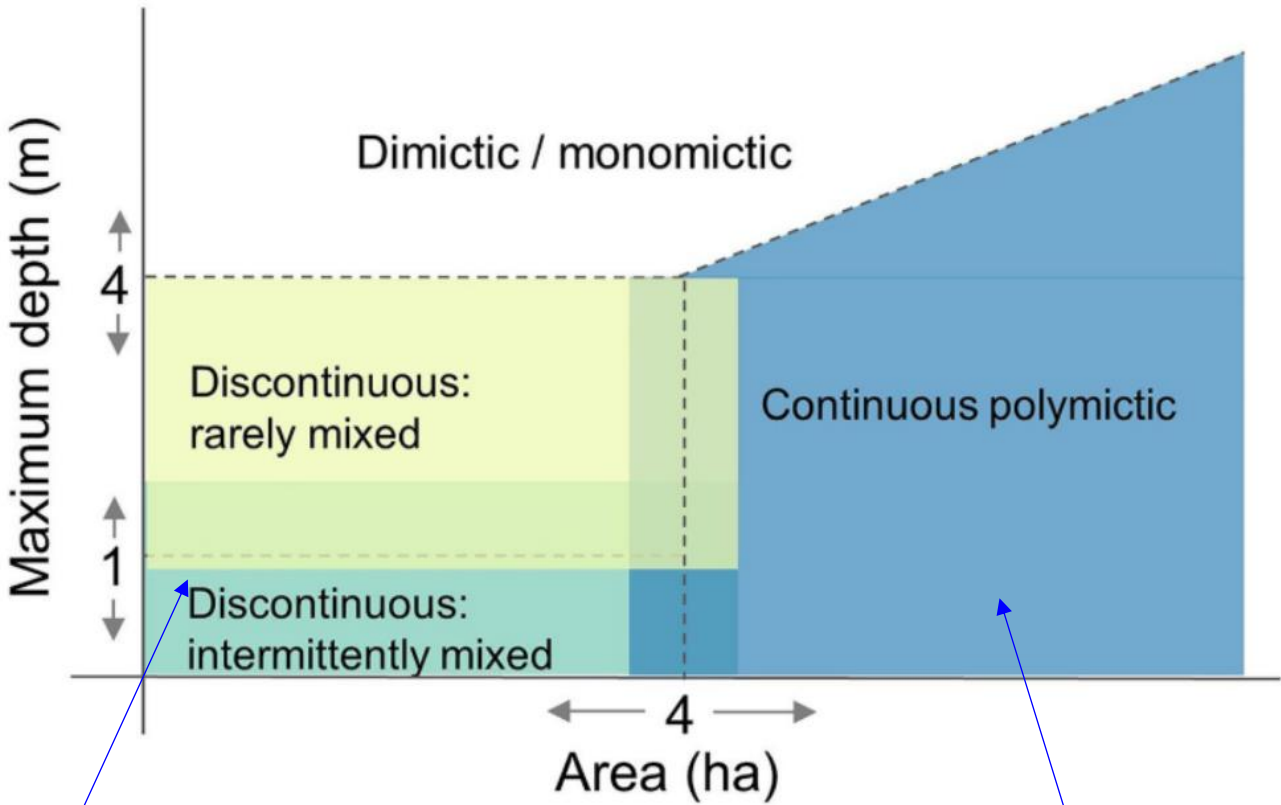
What are ponds and why are they important?

Occurrence of Mixed Conditions:



Classifying Mixing Regimes in Ponds and Shallow Lakes

Meredith A. Holgerson¹ , David C. Richardson² , Joseph Roith³, Lauren E. Bortolotti⁴ , Kerri Finlay⁵ , Daniel J. Hornbach⁶ , Kshitij Gurung³, Andrew Ness³, Mikkel R. Andersen⁷



Mixing Controlled by Heat Fluxes

Mixing Controlled by Wind

Physical limnology of ponds

RESEARCH TRAINING GROUP

SYSTEMLINK

Moving beyond

Aquatic–Terrestrial

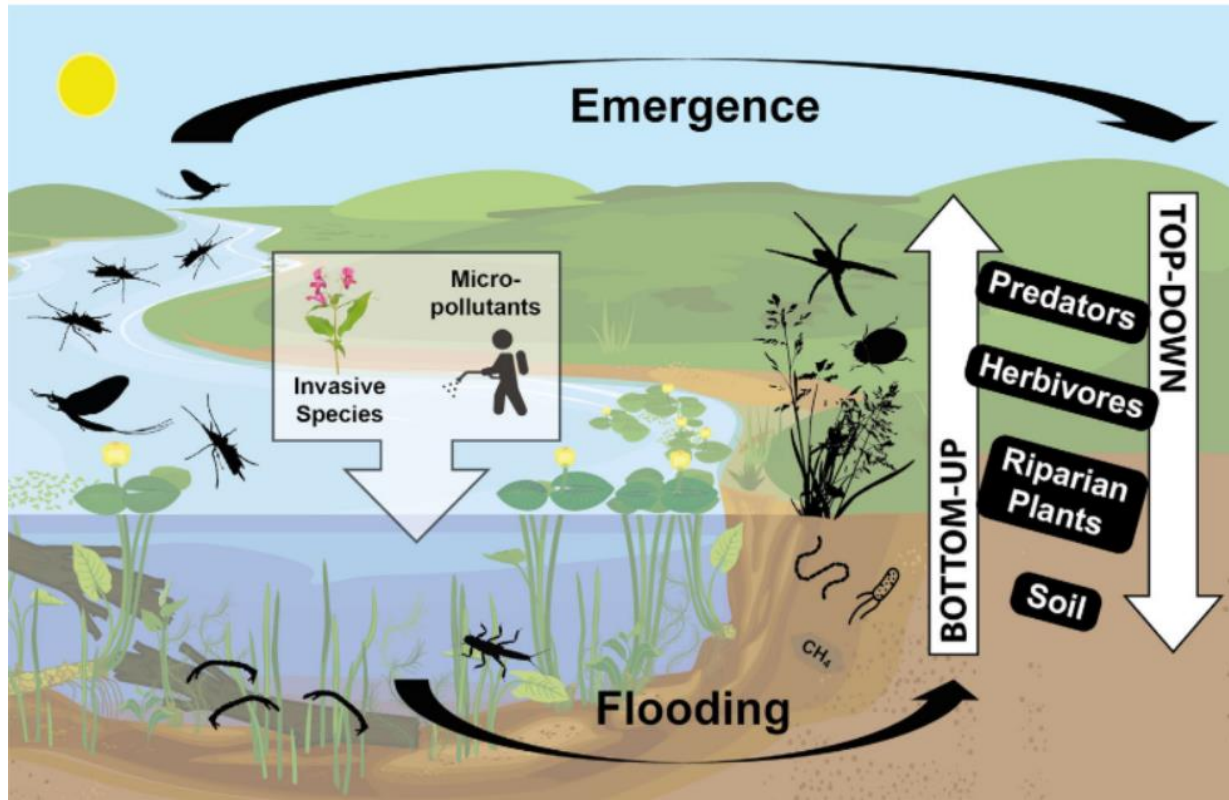
Interactions to Incorporate

Food Web Studies

- 12 replicated floodplain pond mesocosms
- Max. water depth: 0.3 – 0.5 m
- Hydraulically isolated (no groundwater)



Ecosystem Research Facility Eußerthal (EERES, RPTU)



Physical limnology of ponds



Ecosystem Research Facility Eußerthal (EERES, RPTU)

Does biocide treatment for mosquito control alter carbon dynamics in floodplain ponds?

February 2023 · [Science of The Total Environment](#) 872:161978

DOI: [10.1016/j.scitotenv.2023.161978](#)

Biocide treatment for mosquito control increases CH₄ emissions in floodplain pond mesocosms

September 2022 · [Frontiers in Water](#)

DOI: [10.3389/frwa.2022.996898](#)

Benthic macroinvertebrate community shifts based on Bti-induced chironomid reduction also decrease Odonata emergence

October 2022 · [Environmental Pollution](#) 316(2):120488

DOI: [10.1016/j.envpol.2022.120488](#)

Impact across ecosystem boundaries – Does Bti application change quality and composition of the diet of riparian spiders?

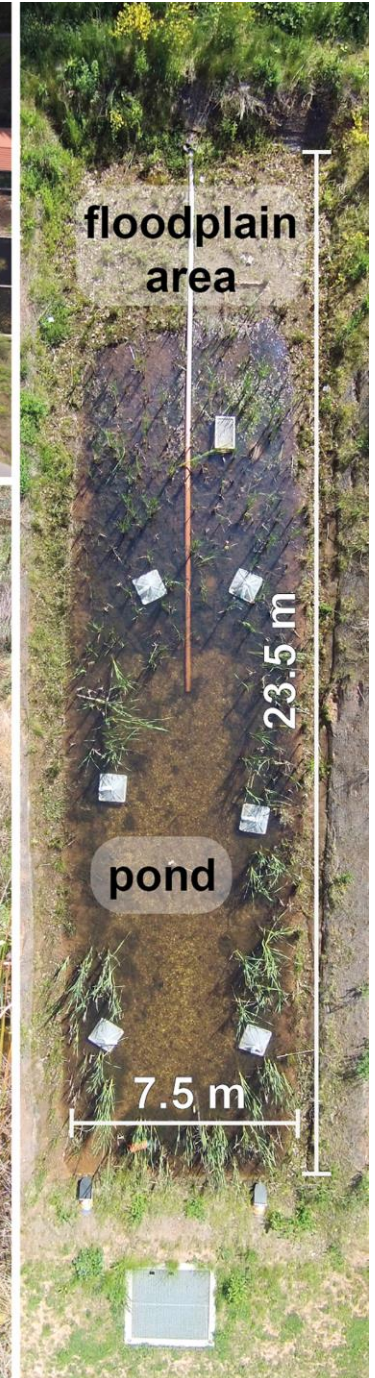
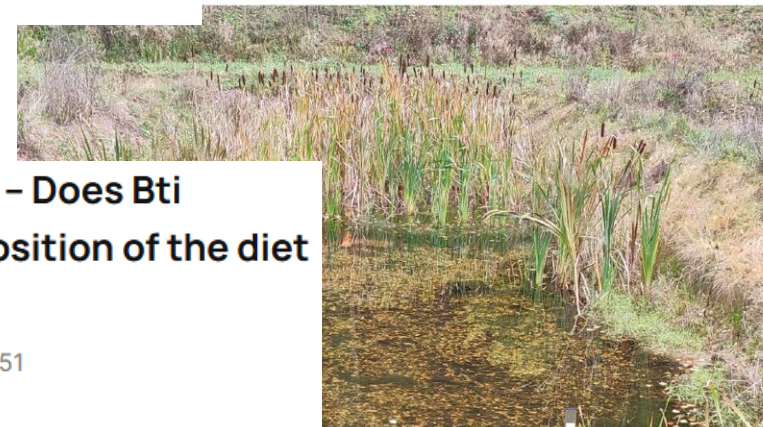
February 2023 · [Science of The Total Environment](#) 873:162351

DOI: [10.1016/j.scitotenv.2023.162351](#)

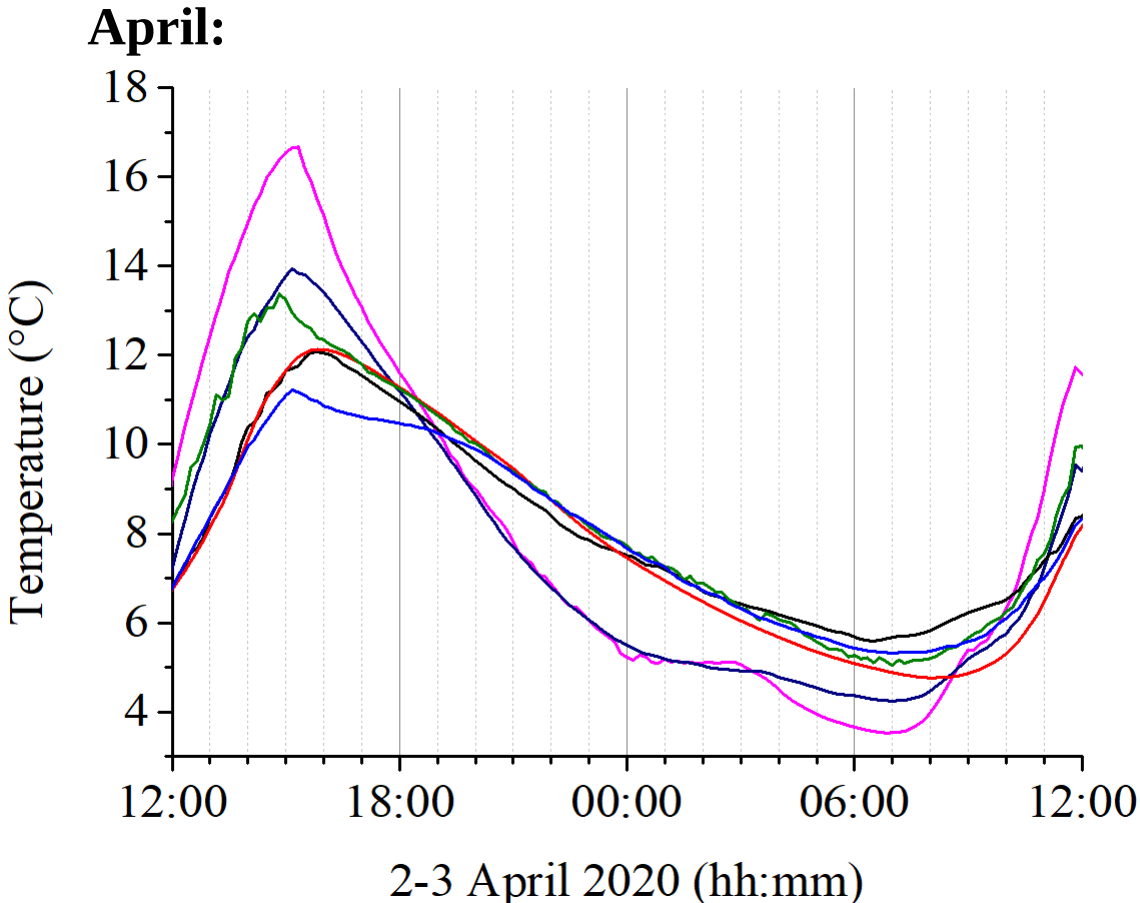
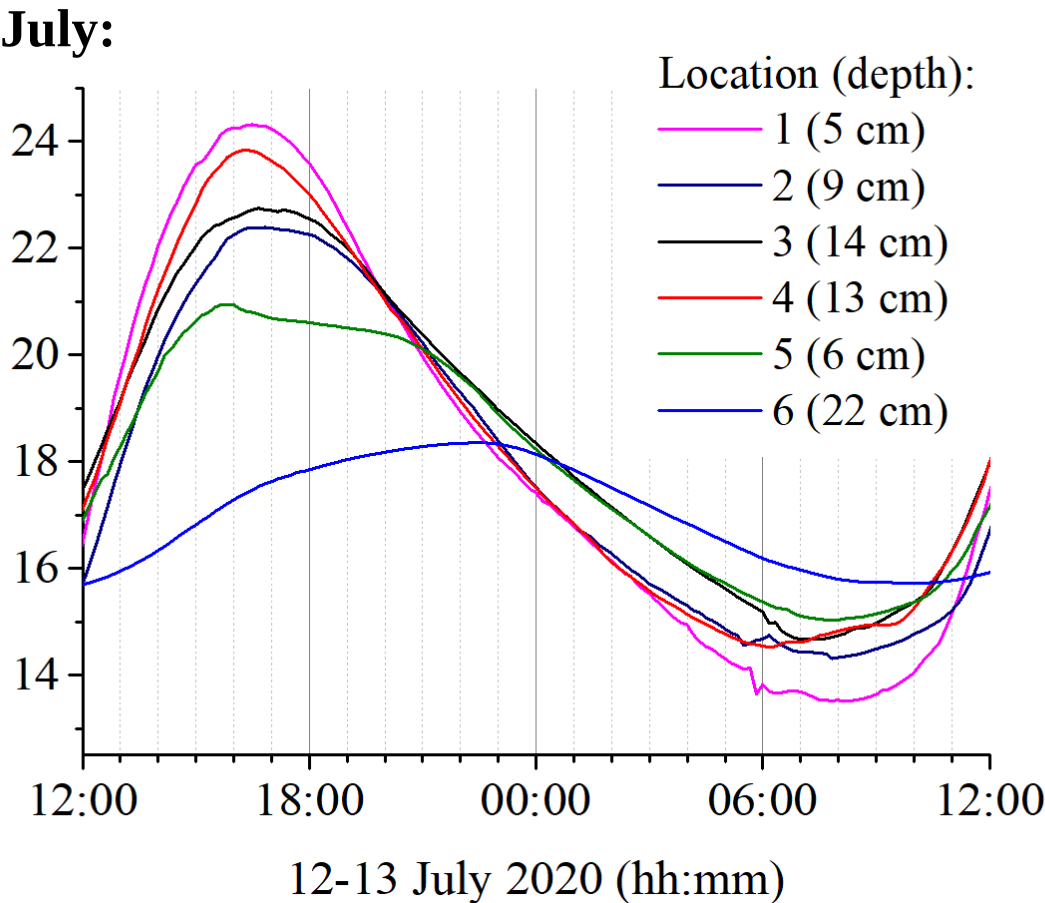
Temperature was monitored at a single location in each pond (deepest point)...

→ What is the spatial and temporal dynamics of temperature, and of horizontal and vertical mixing in the ponds?

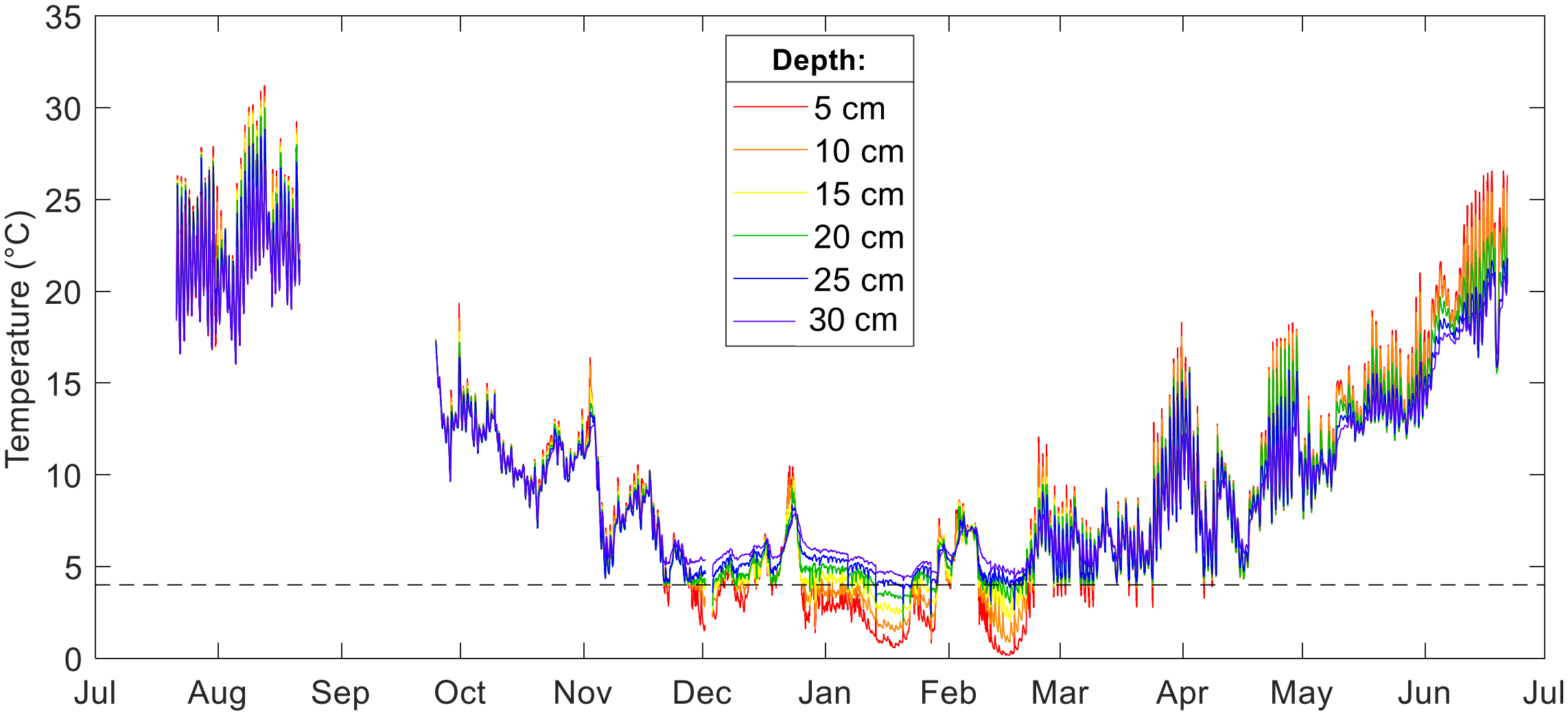
→ How replicable are the physical conditions in the ponds?



Some observations: Spatial variations of temperature

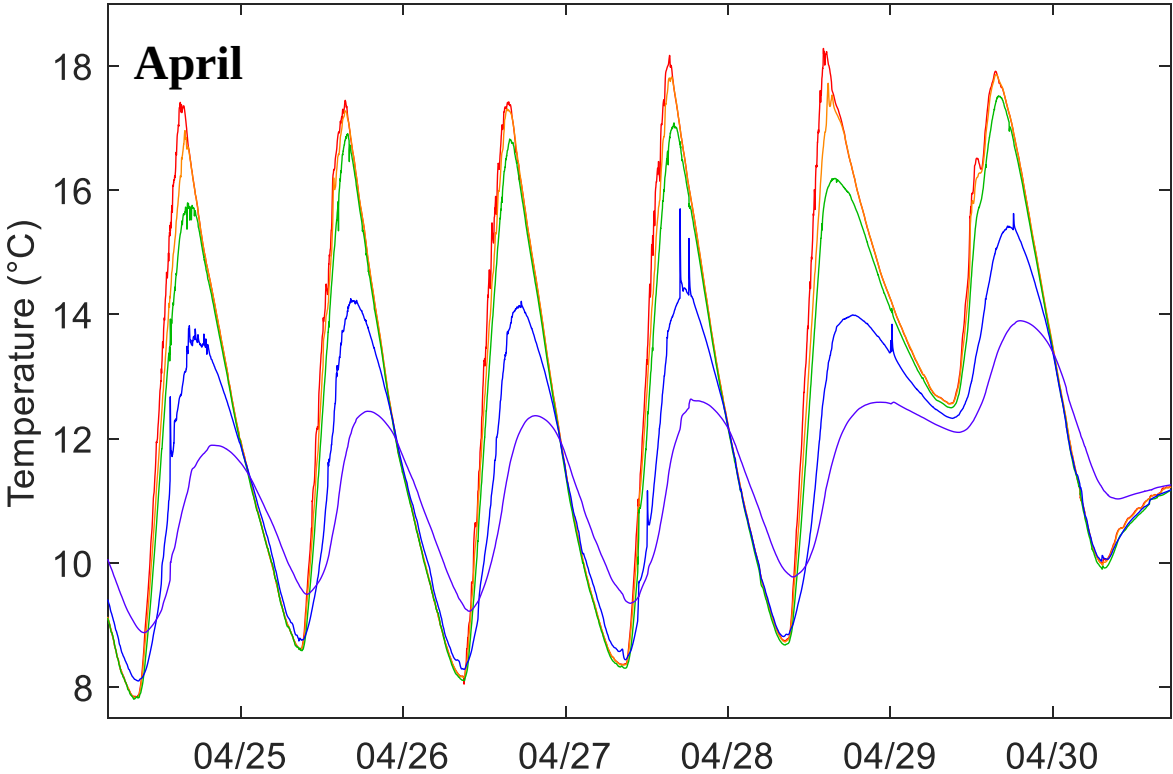
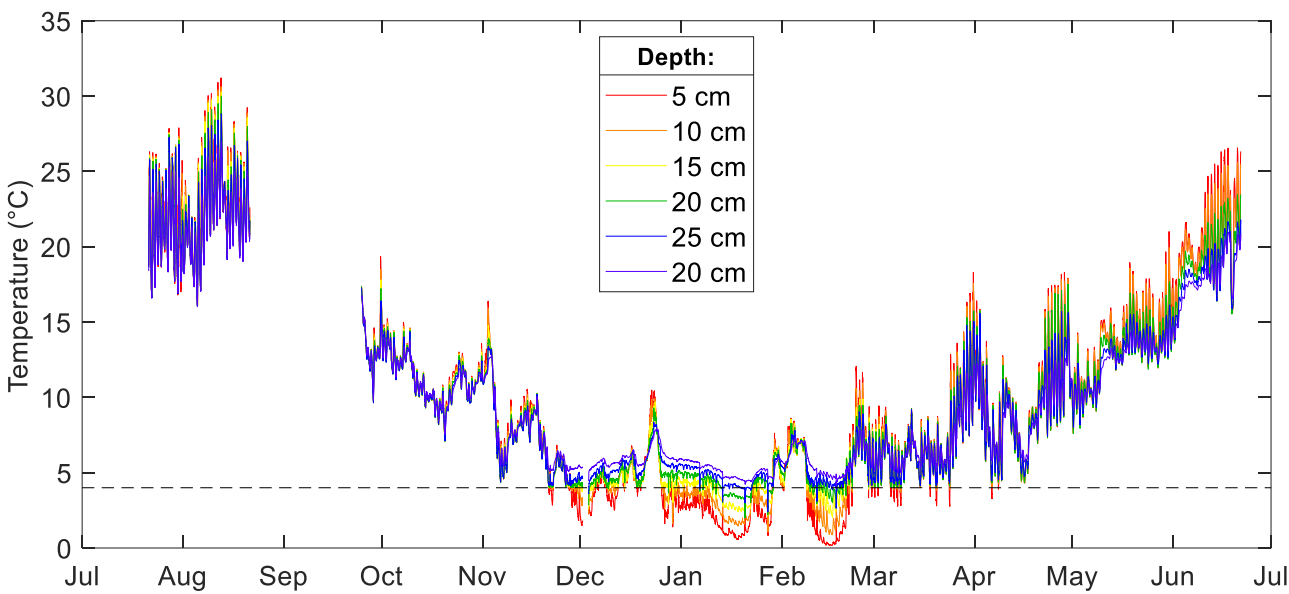
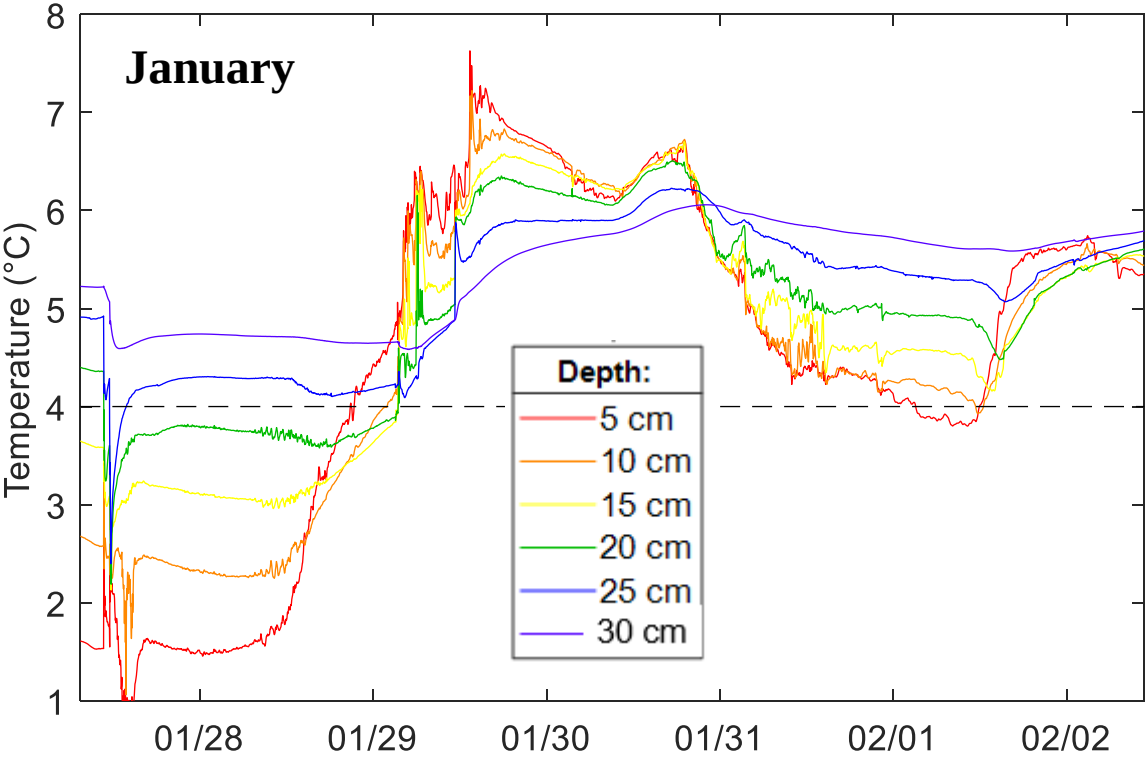


Some observations: Vertical Stratification



Some observations: Vertical Stratification

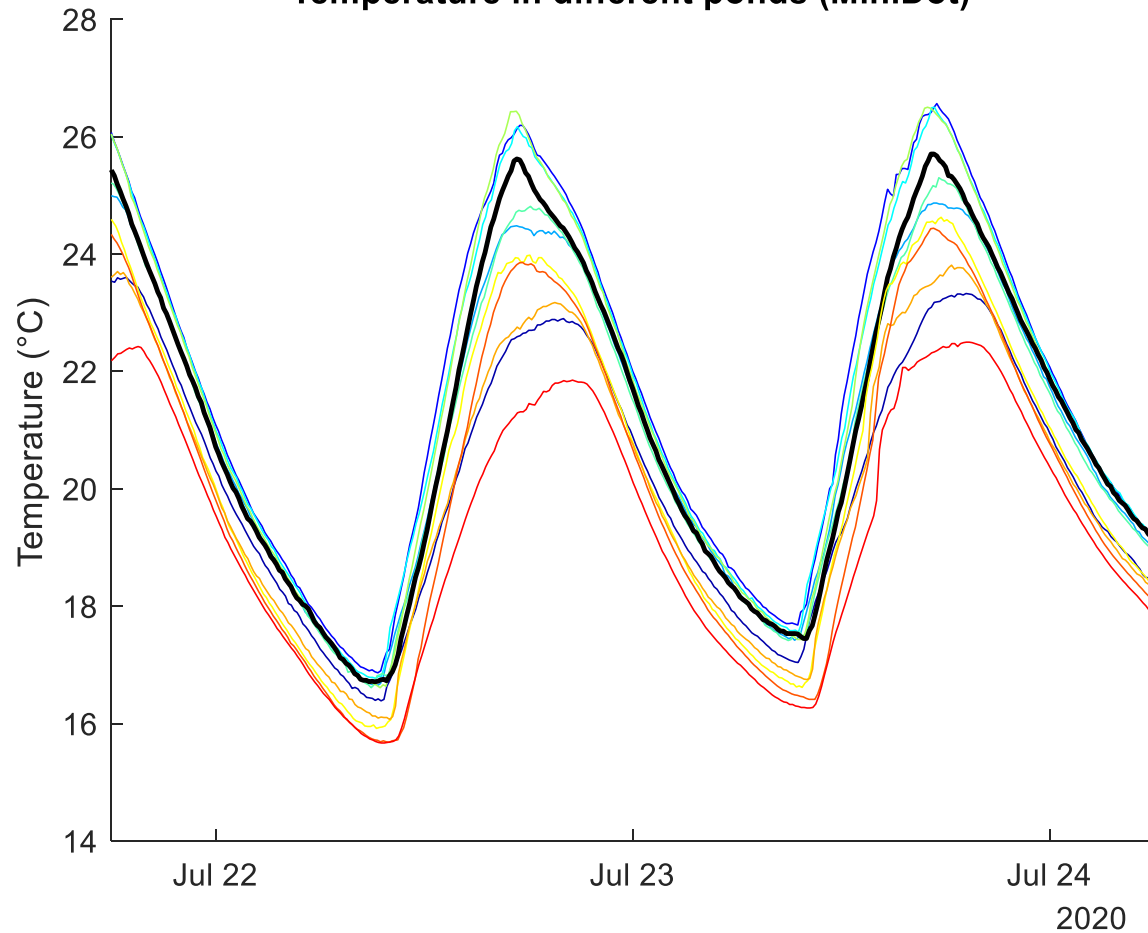
Indications of chemical stratification during winter and spring



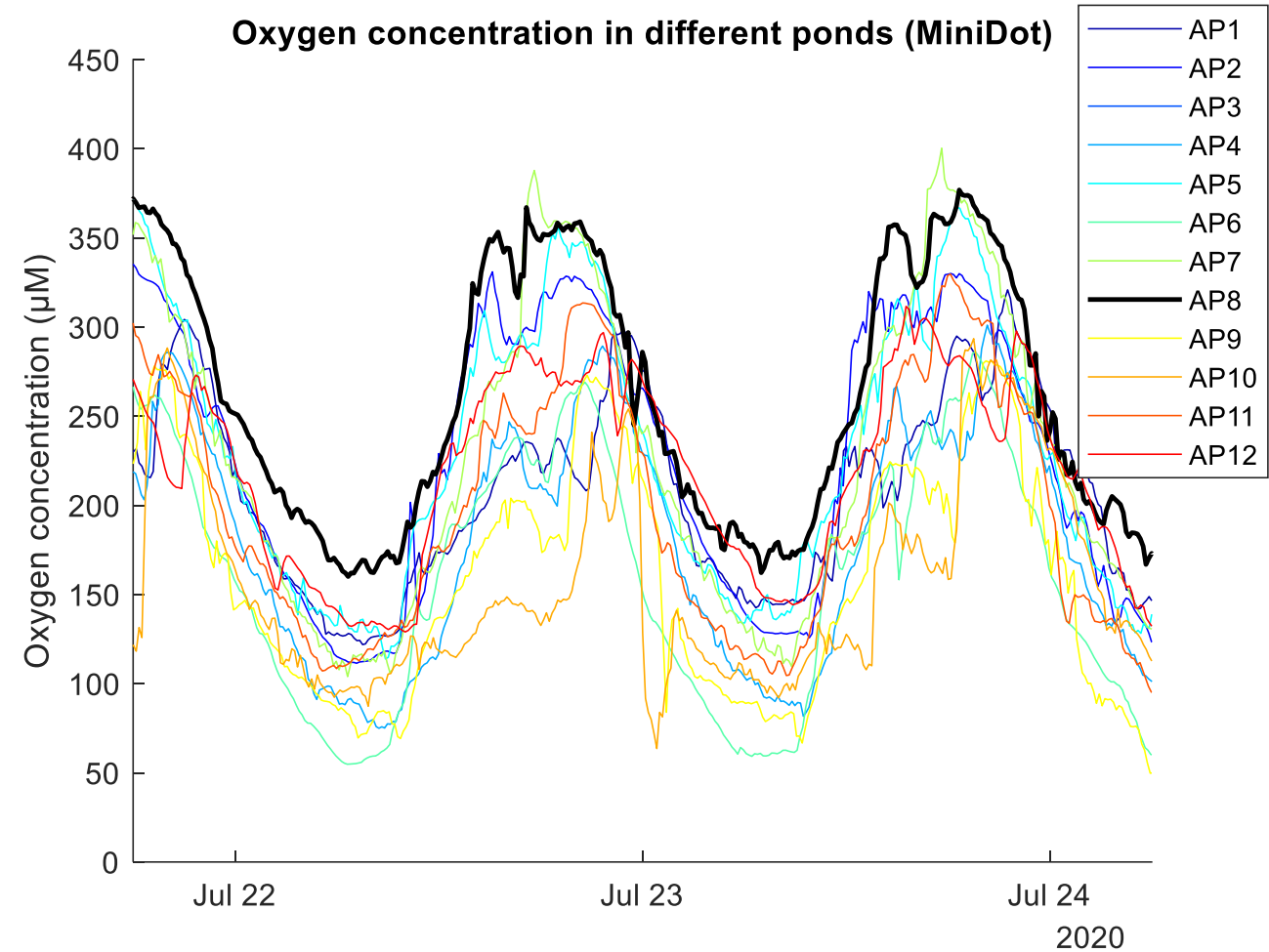
Some observations : Variations among Ponds



Temperature in different ponds (MiniDot)

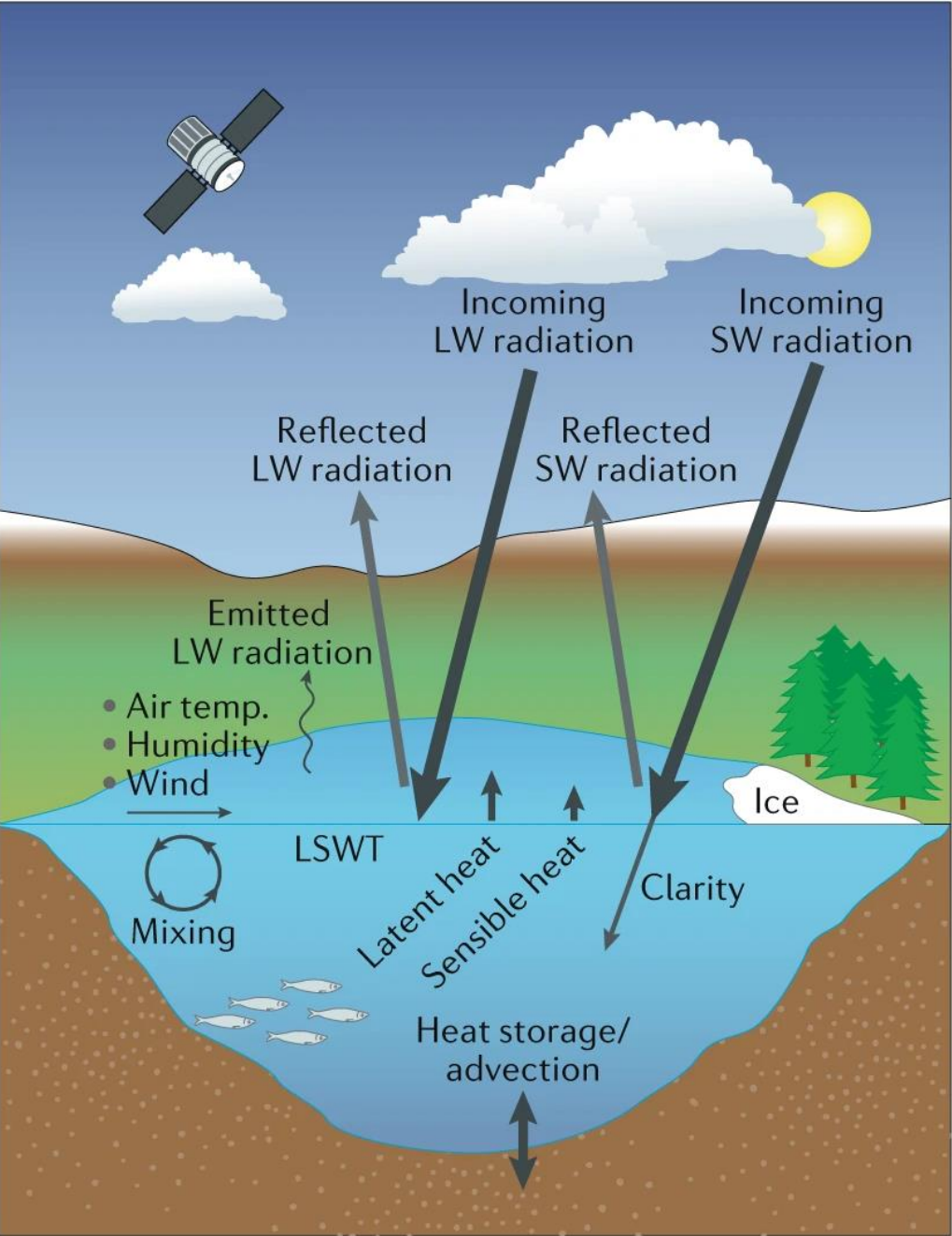


Oxygen concentration in different ponds (MiniDot)

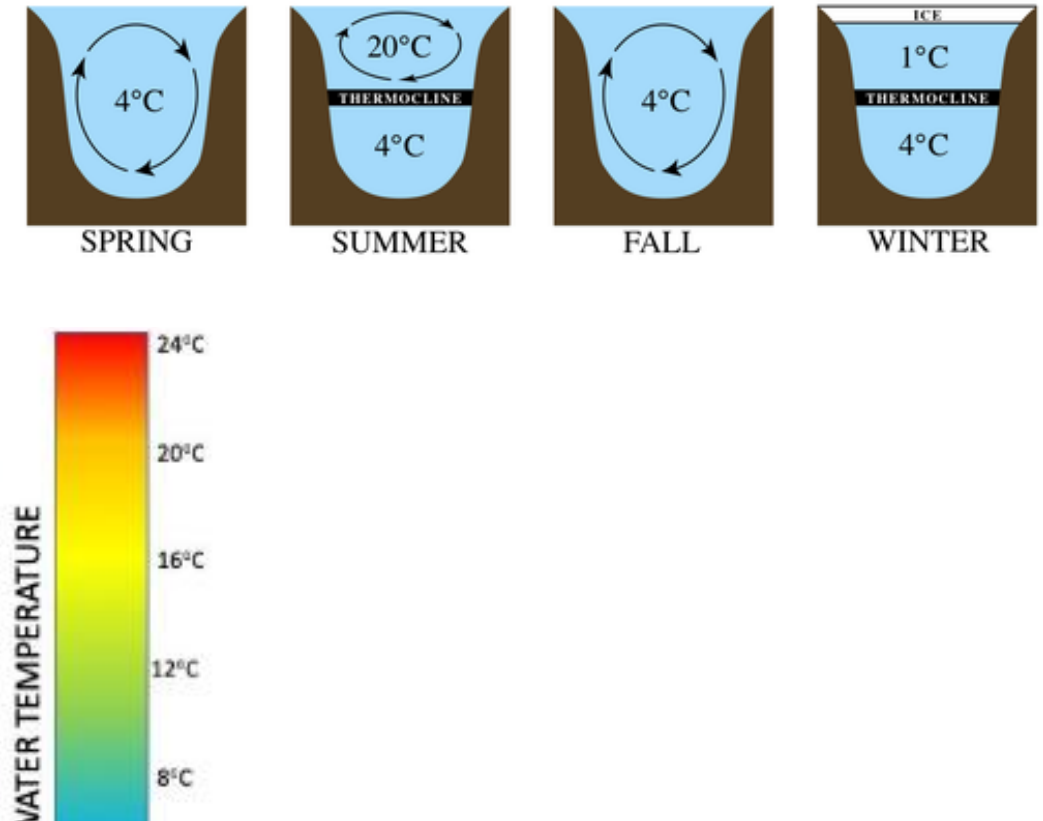
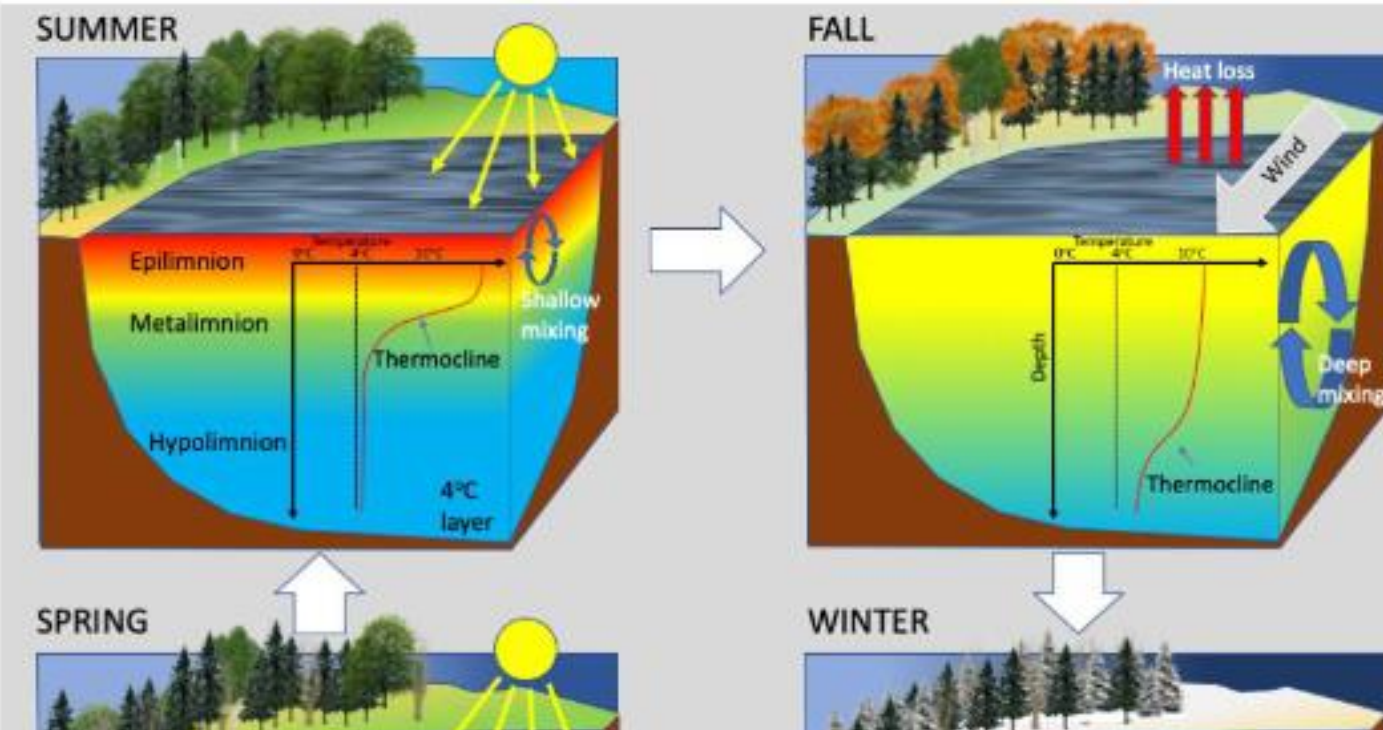


- large variability in temperature and oxygen concentration among different ponds
- effect of different sensor depths / cleaning status or real variability?

Heat Fluxes and Water Temperature:

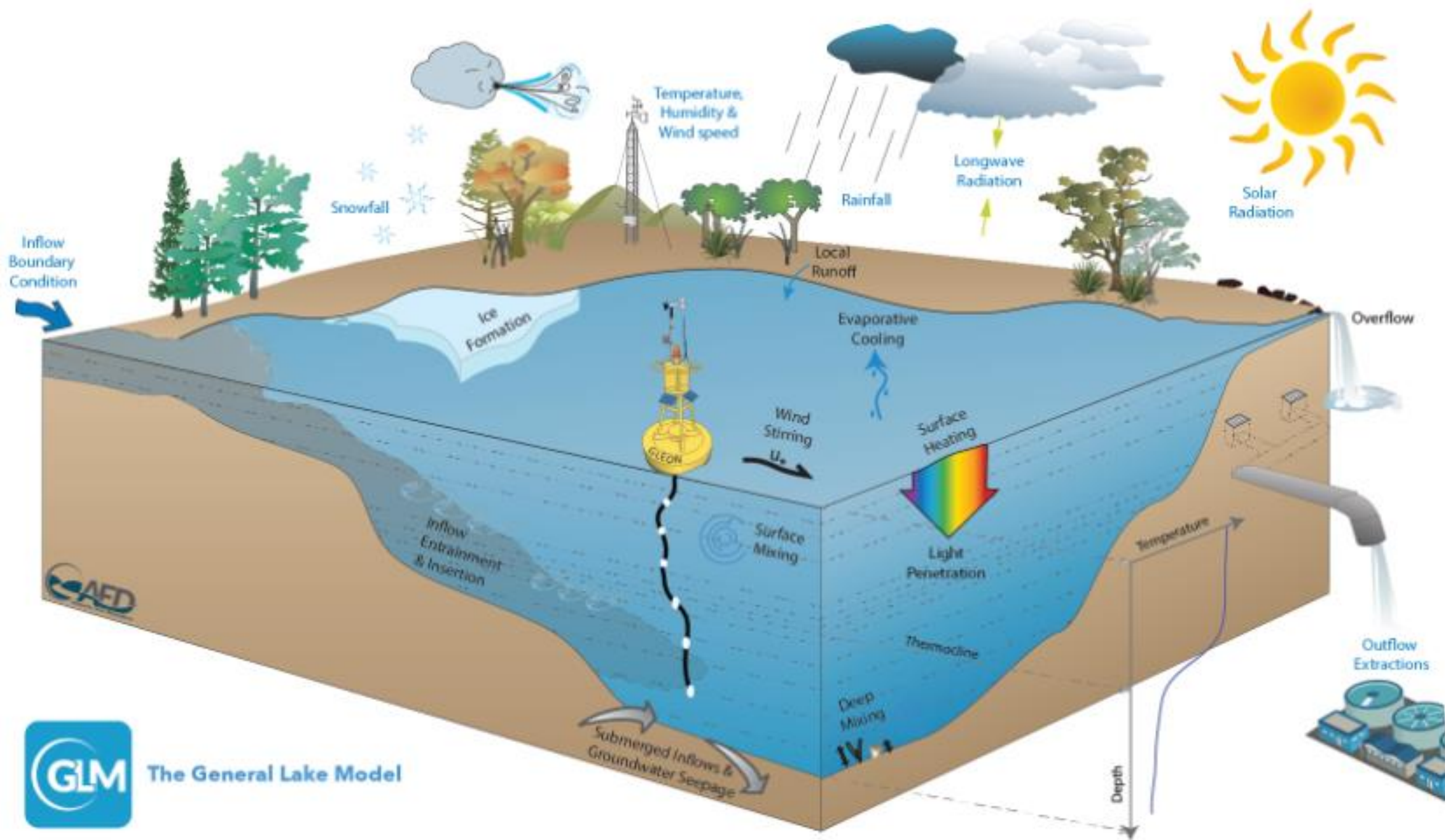


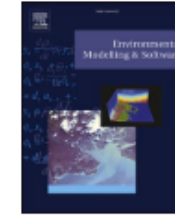
Lake Stratification:



Six common lake types are:

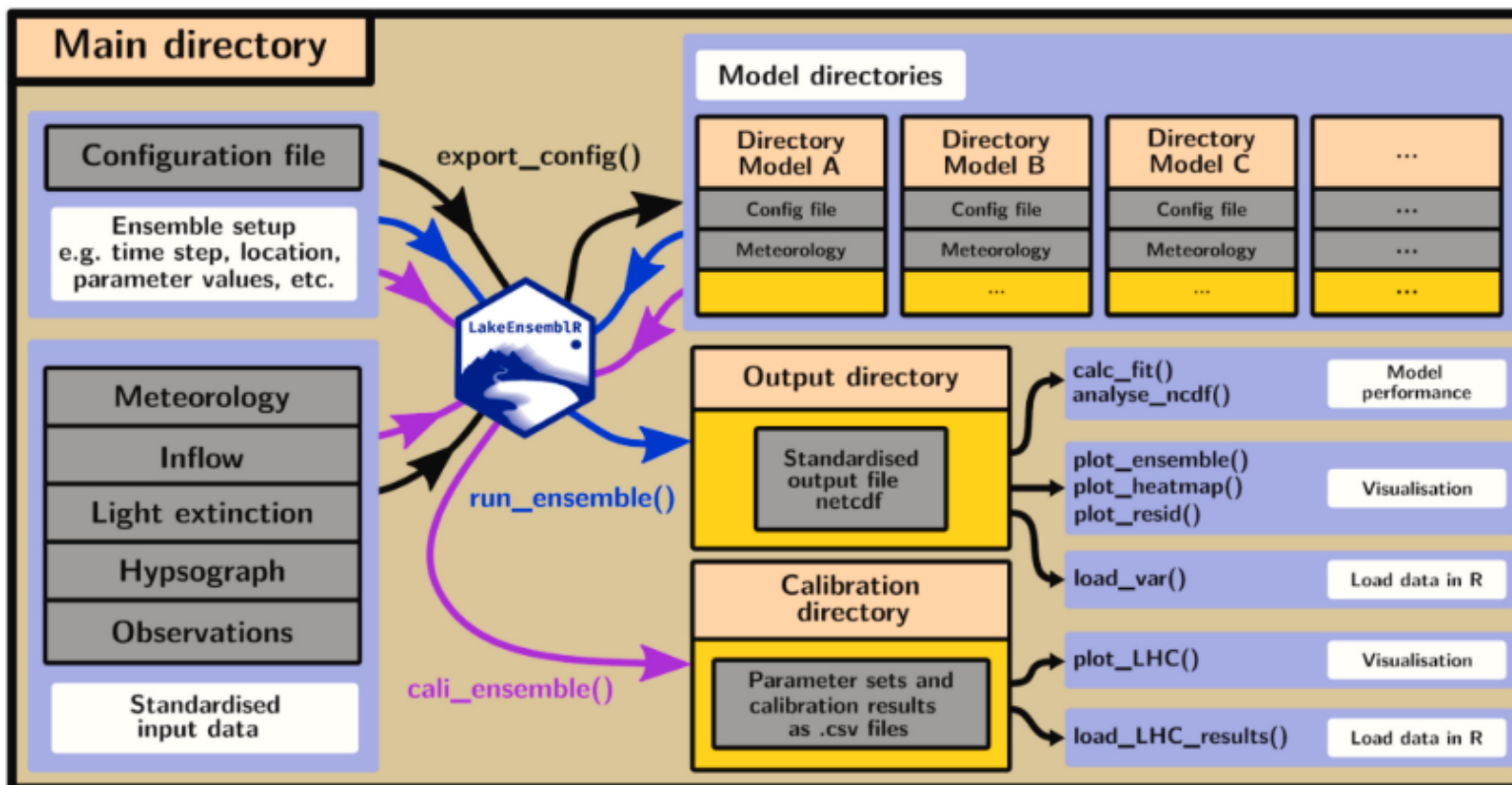
- **Amictic:** covered in ice all year. This mixing pattern is typical of lakes in the Antarctic.
- **Cold monomictic:** water never gets warmer than 4°C and turnover occurs once in summer. Lakes in the Arctic are often cold monomictic.
- **Dimictic:** like at IISD-ELA, lakes are stratified in summer and winter, and mix once in spring and once in fall.
- **Warm monomictic:** water cools to near 4°C in winter, when turnover may occur. Lakes are stratified during other times of the year and are not ice covered. Lakes in the tropics or low altitudes in northern temperate areas can be warm monomictic.
- **Oligomictic:** water generally warm and stratified, occasionally cools to generate circulation at irregular intervals. This mixing type can occur in lower latitude regions where there is little variation in annual temperature.
- **Polymictic:** frequent mixing throughout the year. This type of mixing can occur in lakes across the world and depends on characteristics like depth, size, and shape of a lake.





LakeEnsemblR: An R package that facilitates ensemble modelling of lakes

Tadhg N. Moore^{a,1,*}, Jorrit P. Mesman^{b,c,**}, Robert Ladwig^{d,***}, Johannes Feldbauer^{e,****}, Freya Olsson^f, Rachel M. Pilla^g, Tom Shatwell^h, Jason J. Venkiteswaranⁱ, Austin D. Delany^d, Hilary Dugan^d, Kevin C. Rose^j, Jordan S. Read^k



The minimum data requirements to run LakeEnsemblR are:

- a hypsographic file,
- a light extinction coefficient,
- an initial temperature profile,
- time-series of meteorological forcing variables.



LakeEnsemblR: An R package that facilitates ensemble modelling of lakes

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You can work in groups focusing on the following group-specific objectives:

1. Model validation using observed water temperature: what systematic and random discrepancies exist between models and measurements?
2. Simulation of water temperature in scaled basins (original size of the mesocosms, x10, x100): What is the influence of basin size on water temperature and thermal stratification dynamics?
3. Analyze inter-annual variability (simulations with meteorological data from several years, or data from a weather generator): What is the influence inter-annual variations in meteorological conditions on the seasonal dynamics of temperature and thermal stratification in the ponds?
4. Simulation of climate change scenarios: How will water temperature and thermal stratification change in future climates?

Deadline for portfolio submission: **28.3.2025**

Meetings: on demand. Do you prefer fixed appointments?

Data und documents: <https://seafile.rlp.net/d/e9ec4043ca25495bb8d0/>